

Missing Persons: The Case of National AI Strategies

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Executive Summary

Governance requires trust. If policy makers inform, consult and involve citizens in decisions, policy makers are likely to build trust in their efforts. Public participation is particularly important as policy makers seek to govern data-driven technologies such as artificial intelligence (AI). Although many users rely on AI systems, they do not understand how these systems use their data to make predictions and recommendations that can affect their daily lives. Over time, if they see their data being misused, users may learn to distrust both the system and how policy makers regulate them. Hence, it seems logical that policy makers would make an extra effort to inform and consult their citizens about how to govern AI systems.

This paper examines whether officials informed and consulted their citizens as they developed a key aspect of AI policy — national AI strategies. According to the Organisation for Economic Co-operation and Development (OECD), such strategies articulate how the government sees the role of AI in the country and its contribution to the country's social and economic development. They also set priorities for public investment in AI and delineate research and innovation priorities. Most high-middle-income and high-income nations have drafted such strategies. Building on a data set of 68 countries and the European Union, qualitative methods were used to examine whether, how and when governments engaged with their citizens on their AI strategies and whether they were responsive to public comment.

The authors did not find any country which modeled responsive democratic decision making in which policy makers invited public comment, reviewed these comments and made changes in a collaborative manner. As of October 2022, some 43 of the 68 nations and the EU sample had an AI strategy, but only 18 nations attempted to engage their citizens in the strategy's development. Moreover, only 13 of these nations issued an open invitation for public comment and only four of these 13 provided evidence that public inputs helped shape the final text. Few governments made efforts to encourage their citizens to provide such feedback. As a result, in many nations, policy makers received relatively few comments. The individuals who did comment were generally knowledgeable about AI, while the general public barely participated. Policy makers are therefore missing an opportunity to build trust in AI by not using this process to involve a broader cross-section of their constituents.

Introduction

The world's people are at a crossroads in governing for AI. AI can be defined as a machine-based system that can, for a given set of human-defined objectives, make predictions, recommendations or decisions influencing real or virtual environments.¹ AI systems are often global and demand for AI products and services is growing among business, government and civil society.²

AI systems hold enormous potential to enhance human capacity, increase productivity, catalyze innovation and help mitigate complex problems. However, public concern about AI systems is on the rise.³ AI systems are often designed and deployed in an opaque manner that many users cannot see. Moreover, individuals may struggle to understand how these systems make decisions and thus, they are unlikely to trust these processes. If policy makers want to encourage continued development and use of these systems, these same officials have a responsibility to inform, consult and involve their citizens about how AI is designed, developed and deployed.

Trust is situational and relational and not easy to define. Scholars agree that trust underpins all human contacts and institutional interactions. Moreover, they note that once trust is lost or eroded, it is not easy to regain or sustain (OECD 2022; Kumagai and Iorio 2021). AI deployers, such as AI policy makers, therefore have a stake in ensuring that AI is trustworthy (The Future Society and EYQ 2022).

This paper examines if governments inform and involve their citizens as they develop a key aspect of AI policy — national AI strategies. Although the OECD tracks such strategies as part of its efforts to encourage trustworthy AI, the OECD does not explicitly define what constitutes an AI strategy. The OECD does however delineate the objective of such strategies. On its website, OECD.AI, it notes such strategies articulate a government's vision regarding AI's contribution to the country's social and economic development. These strategies set priorities for public investment, identify what research taxpayers should fund and what regulatory steps policy makers should take.⁴

The OECD notes that governments often involve their stakeholders to obtain input on the design of their national AI policies and strategies. "Public consultations leverage different tools including interviews, surveys, online discussion fora and events such as hearings,

¹ The OECD's definition has been used because it is internationally accepted (OECD 2019).

² According to Fortune Business Insights (2022), the global AI market size is projected to grow from US\$387.45 billion in 2022 to US\$ 1394.30 billion in 2029.

³ As an example of public concerns, see Rainie et al. (2022); while few experts see ethical AI adoption as a comprehensive solution, see Rainie, Anderson and Vogels (2021).

⁴ See https://oecd.ai/en/dashboards/policy-instruments/National_strategies_agendas_and_plans.

workshops, seminars, focus groups and conferences....Expert consultations usually help define the issues, formulate policy objectives and, in some cases, assess policy effectiveness. In addition to expert consultations, countries such as Canada or Chile engage citizens to ensure that a diverse range of perspectives is considered” (Galindo, Perset and Sheeka 2021, 7). Consequently, government AI strategies are multidimensional and reflect compromise among a wide range of actors inside and outside government (Osborne and Plastrik 1997).

Nations take different approaches to these strategies, as shown by various attempts to map and compare them (Dutton 2018; Struett 2019; 2020; Saran, Natarajan and Srikumar 2018; Tortoise Media n.d.; Fjeld and Nagy 2020; Stanford 2021, 5). For example, Singapore’s AI strategy aims to “identify areas to focus attention and resources on at a national level; set out how the Government, companies and researchers can work together to realize the positive impact from AI, and address areas where attention is needed to manage change and/or manage new forms of risks that emerge when AI becomes more pervasive.”⁵ In contrast, the United Kingdom’s AI strategy aims to: “1. Invest and plan for the long-term needs of the AI ecosystem to continue our leadership as a science and AI superpower; 2. Support the transition to an AI-enabled economy, capturing the benefits of innovation in the UK, and ensuring AI benefits all sectors and regions; and 3. Ensure the UK gets the national and international governance of AI technologies right to encourage innovation, investment, and protect the public and our fundamental values.”⁶

In 2021, the authors, as staff at the Digital Trade and Data Governance Hub at George Washington University, developed a metric of data governance around the world. The metric (as of December 2022) covers 68 countries and the European Union, and examines how nations govern three types of data — public, personal and proprietary data.⁷ AI strategies are one of the 26 indicators of data governance, and it became the key source of data to begin the research. In describing this indicator, the Hub noted: “AI strategies outline a national vision for how a nation can build and/or maintain its ability to create and utilize AI for commercial as well as societal use. They often provide guidance to government agencies, discuss investments in AI research and development, and discuss the role of government in developing standards and the rule of law for this emerging technology” (Zable, Struett and Aaronson 2022, 4).

Building on previous efforts to map data, the authors examined whether, when and how governments engaged their publics in the development of AI strategies, and then to

⁵ See www.smartnation.gov.sg/initiatives/artificial-intelligence/.

⁶ See www.gov.uk/government/publications/national-ai-strategy/national-ai-strategy-html-version.

⁷ The metric includes six attributes of data governance: strategies; laws and regulations; structural changes (has the government created new bodies or tasked individuals/groups to monitor or enforce data governance); human rights and ethical guidelines; whether they consult their public in at least one area of data governance; and mechanisms for international cooperation.

identify which citizens participated. They also also whether governments developed inclusive processes to seek public comment and if governments responded to citizen concerns They frequently found the public was missing from decision-making.

As of October 2022, some 43 of the 68 nation and EU sample had an AI strategy, but only 18 attempted to engage their citizens in the strategy's development. Moreover, only 13 of these nations issued an open invitation for public comment. Only four countries — Chile, Germany, the United States and Uruguay — indicated how they changed their strategies in response to public comments. Although some acknowledged the comments they received, most governments did not make changes in response to the comments that they received. In most nations, moreover, few people commented and those who did were knowledgeable about AI and willing and able to articulate their concerns. Hence, policy makers are missing an opportunity to build trust in AI by not using this process to involve a broader cross-section of their constituents. AI governance may be aimed at the people, but it is not by the people.

AI, Trust and Governance

AI has become a part of daily life for many users. They interact with AI systems as they work, shop, learn and seek companionship. In recent years, AI systems have become so humanlike that, in many instances, users do not know if they are interacting with an AI such as a bot (PEGA 2017). Even the people who design AI systems may not understand how that algorithm makes predictions, recommendations or decisions (Rainie et al. 2022; Ammanath n.d; Li et al. 2022) , because AI is developed using algorithms that create an opaque decision tree (Rudin and Radin 2019). Not surprisingly, AI may seem untrustworthy to members of the general public.

As with every innovation, AI systems reflect the normative judgments of those who design and develop the systems. Until recently, few developers and designers were trained to create systems that can effectively weigh ethical and moral factors as they make predictions, recommendations or decisions.⁸ But AI systems are unable to consider context or nuance, or more broadly, to effectively make value-based judgments (Kapoor and Narayanan 2022). Moreover, users and employers cannot see how designers and developers incorporate normative judgements. Because of this opacity, designers, developers and employers must develop ways to show the public that their systems are

⁸ <https://www.cs.utexas.edu/news/2022/importance-ethics-computer-science>; <https://news.stanford.edu/report/2022/06/28/new-program-embeds-ethics-computer-science-courses/> and Stavrakis et al: 2022.

reliable, accountable and trustworthy, and these systems must exhibit and sustain trustworthy behaviour if designers and deployers want end users to accept AI. Thus, growing numbers of people involved in the development of AI now believe that public involvement in the design, deployment and governance of AI is essential, because it can give citizens a voice and a measure of control over AI systems. Without such a feedback loop, society is unlikely to accept AI (Stanton and Jenson 2021; Birhan et al. 2022).⁹

⁹ Both the US National Institute of Standards and Technology (NIST) and the NSF support research in participatory AI. See <https://beta.nsf.gov/funding/opportunities/nsf-program-fairness-artificial-intelligence>; and <https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&cad=rja&uact=8&ved=2ahUKEwjmwMGd5OaAAxX5GFkFHT6NDUYQFnoECBAQAQ&url=https%3A%2F%2Fwww.nist.gov%2Fnews-events%2Fnews%2F2023%2F05%2Fnist-partners-nsf-new-institute-trustworthy-ai-law-society-trails&usg=AOvVaw2HLfuP8BCMZ1YmzSf0-fci&opi=89978449> NIST (2022; n.d.) and <https://oecd.ai/en/catalogue/tools/participatory-ai-framework>

Box 1: What Is Trustworthy AI?

It was not possible to find a widely accepted definition for “trustworthy AI” because trust is a normative concept. Policies designed to inspire trust in one country may not yield trust in another. The OECD has, however, tried to come up with an internationally accepted definition — it defines trustworthy AI as “AI systems that embody the OECD AI Principles; that is, AI systems that respect human rights and privacy; are fair, transparent, explainable, robust, secure, and safe; and the actors involved in their development and use remain accountable.” The Chinese government has put forward a slightly different definition; although it has committed to adhere to the OECD AI Principles, AI trustworthiness “reflects the trustworthiness of AI systems, products, and services in terms of security, reliability, explainability, and accountability. Trustworthy AI implements ethical governance requirements from the perspective of technology and engineering practice to achieve an effective balance between innovative development and risk governance.”

Sources: (OECD 2021, 6); China Academy of Information and Communications Technology [CAICT] and JD Explore Academy 2021, 4–5).

Meanwhile, citizens expect government officials to design public policies that allow society to reap the benefits of AI while simultaneously protecting users from harm.¹⁰ In recent years, policy makers have created a diverse set of national and international initiatives to ensure trustworthy AI, ranging from shared principles to regulations (Turner Lee and Lai 2022; Djefal 2021).

Policy makers are responding to these concerns about trust and AI for many reasons. First, they understand that AI systems are now essential to national security,¹¹ a key responsibility for all governments (International Telecommunications Union 2018). Second, they understand how AI could yield prosperity. AI underpins other emergent technologies such as virtual reality, while firm investments in AI can improve productivity and

¹⁰ See Tschopp and Quadroni (2022).

¹¹ It is so important to the members of the North Atlantic Treaty Organization (NATO) that they too created a strategy for AI, which they call autonomous systems. See NATO (2022).

innovation. Nations need AI capacity to reap the benefits of data analytics and other forms of data-driven change (Yang 2022; Cockburn, Henderson and Stern 2018). Policy makers also recognize that AI also holds great promise to help mitigate wicked problems such as climate change (Aaronson 2020).

But policy makers' actions to promote AI can also undermine AI and trust in government. Government officials that misuse or allow firms to misuse these systems can, with or without intent, undermine human rights, particularly those of marginalized individuals and communities (Heikkilä 2022). Researchers have found that these groups often face disproportionate harms and discrimination from AI systems (see, for example, Birhane et al. 2022; Borgesius 2018). Public concern about these negative effects is leading to distrust in AI (Bryson 2022). Not surprisingly, influential groups in business, government and civil society are demanding policy makers take steps to build and sustain trust in AI (Porway 2021).¹²

Policy makers can build trust in governance by collaborating with their own constituents on solutions or mitigating strategies to the many problems they confront. But citizen engagement is not easy. In democracies, citizens are simultaneously economic, political and social actors and, as such, tend to use their limited time, energy and voice on a small range of issues they care about (Olsen 1971). AI may not be one of those issues, because of its complexity and opaque nature. However, policy makers in democracies need the blessing of these undermotivated citizens to remain legitimate, which can lead to a "catch-22." As the World Bank notes, "Without citizens' trust in government, formal citizen engagement is unlikely. Without citizens' participation, government's performance will be poor, and trust in government will fall" (Kumagai and Iorio 2020, 14).

Finally, the authors noted that there are many additional benefits to public consultation on complicated issues such as AI. First, the broader public often sees issues from a different angle and may provide new insights to policy makers. Moreover, by consulting a broad swathe of its citizenry, the nation may increase regulatory literacy, which, in turn, may yield greater compliance with regulations. Lastly, the feedback loop may ensure that as societal needs and the public interest evolves over time, policy will evolve too (OECD 2011, 9).

Methodology

The Hub sought to examine whether, how and when nations informed and consulted with their citizens about their AI strategies prior to their release. The research strategy was built on the dataset mentioned earlier, which includes 26 indicators of data governance practices for 68 countries and the European Union (a total of 69). One of these indicators

¹² See <https://data.org/news/charting-the-data-for-good-landscape/#responsible-ai-advocates>.

delineates whether a nation has developed and adopted an AI strategy. These 69 data points formed the dataset used for this analysis. The 69 governments in the sample represent a mix of incomes and regions based on the World Bank's categorizations.¹³ While the Hub's mapping does not cover every country with an AI strategy, this paper covers many of the ones listed at the OECD (62) and a preponderance of those with AI strategies in the world. However, it is acknowledged that it is not a representative sample of the world's countries.

Other scholars have examined the role of the public in developing AI strategies. In 2022, researchers at Derechos Digitales focused on the process in Brazil, Chile, Colombia and Uruguay, and found that governments tended to rely on online platforms and email to seek out public opinion. They concluded that the consultative processes they reviewed were inadequate because they were not inclusive toward women, historically marginalized or geographically remote communities, citizens with disabilities or those lacking internet access or other resources. Furthermore, the processes were not collaborative, because officials only asked for public input at a late stage of strategy development. Finally, they noted that most of these Latin American countries were unable to promote informed engagement or follow up (Hernández and de Souza 2022, 37). UNICEF has examined whether and why AI strategies ignore the needs of children. UNICEF researchers used a literature review to explain this gap, but they did not examine whether advocates of children or children were consulted (Penagos Kassir and Vosloo n.d.). Finally, Janis Wong et al. (2022) explored the use of AI by public administration. The authors worry that without participatory governance, AI systems can easily be misused and consequently, these systems may fail the people they were supposed to serve.

As these works illuminate, officials that attempt to organize public consultations can face obstacles. It is not always easy to motivate people to participate in consultations that may seem far removed from their day-to-day problems. In addition, government officials may lack the will to incorporate the results into workable policies (Culver and Howe 2003).

This research focuses on the process of consultation. The methodology was based on norms and levels of consultation articulated by the International Association for Political Participation (IAP2). The IAP2 asserts that democracies have shared norms (values) and processes for citizen involvement in democratic decision making. It argues that citizens should be involved in the decision-making process and governments should use their comments as they revise official documents. Moreover, it asserts that policy makers should seek out the public to comment and offer more ways for individuals to participate online

¹³ Six countries from each region were chosen that also represented a mix of high, low and middle incomes. This dataset can be found at: <https://datagovhub.elliott.gwu.edu/research/>. The World Bank's regional and income characterization is available at: <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups>.

and/or in person. Participants should have the information they need to participate in a meaningful way. Finally, policy makers should communicate to the participants how their input affected the decision.¹⁴ The IAP2 has also created a model of levels of participation that range from simply informing the public that they have an AI strategy, to consulting the public, to the highest levels of participation where policy makers collaborate with and empower their constituents.

The investigation began by reviewing the literature and delineating the research questions. For each stage of this research, a qualitative approach based primarily on publicly available information found online was utilized.

The goal was to understand if invitations for public comment inspired a wide range of participants, including non-experts, to provide advice on AI strategies. Following the work of Mancur Olson (1982), a distinction was made between organized and unorganized publics. Olson notes that most of the time, most citizens do not participate directly in governance because they believe their individual decisions and votes can have little influence, so they “are rationally ignorant” about public affairs.

But Olson also stresses that the same individual who is not generally politically active can become motivated on a specific issue of concern to themselves or their family. If they work at a coffee shop with irregular hours they might join a union, or if their child has cancer, they may join the local cancer society. These individuals will likely want that union, civil society group or professional association to influence government on a particular issue of great concern to their ethics or economic situation. This individual is now also a member of the organized public — a group that works to provide its members with important information about issues that can enable the group and its members to thrive (*ibid.*, 26). The organized public includes civil society associations such as Human Rights Watch¹⁵ and the Internet Society, professional associations such as the International Association of Electrical and Electronic Engineers¹⁶ or business associations such as the Computer and Communications Industry Association.¹⁷ They are also referred to as “special interest groups.”

Special interest groups are not alike — some are grassroots organizations, driven by members and reflective of activist member opinion, while others are more staff driven. Yet they play a major role in public policy in democracies (*ibid.*). First, they often hire lobbyists to ensure their interests are heard. Second, representatives of unions, firms and

¹⁴ The IAP2 aims to advance and extend the practice of public participation through professional development, certification, standards of practice, core values, advocacy and key initiatives with strategic partners around the world. See www.iap2.org/page/corevalues.

¹⁵ See www.hrw.org/.

¹⁶ See www.ieee.org/about/ieee-history.html.

¹⁷ See Computer and Communications Industry Association (2022).

professional associations are asked to testify or to join advisory bodies. In so doing, they can develop relationships with parliamentary or legislative staff who often move on to lobbying or policy jobs in such organizations (Center for American Progress 2017). But in general, these organizations are listened to and have more opportunities to be heard by policy makers (Anderson 1991). In contrast, the unorganized public speaks to policy makers only when asked.

Hence, the authors examined consultations involving both the organized public and the unorganized public, and not those open solely to experts. It is difficult, however, to distinguish why a person participated and under what identity because individuals are multidimensional. Person A can simultaneously be an expert on AI as well as a citizen who may not care to comment on the governance of AI. In contrast, person B may be a member of Human Rights Watch who is also an expert on big data and who actively comments on AI governance.

Next, the authors delineated AI strategies as a national statement of the country's vision for AI in the economy (and often in the polity). The objective was to make the definition as broad as possible to include a wide range of strategies governments created to obtain public comment. Therefore the authors included formal as well as informal government consultations. The authors defined a formal consultation as one that was officially published, and which had a verifiable consultation where the public was officially invited to give comments. Next the authors categorized consultations as informal if they lacked an official invitation, such as those releasing a discussion paper without explicit requests or means to provide feedback, or if they sought feedback or provided avenues for contribution only *after* the official strategy's publication. Those that consulted *before* they released an AI strategy were also included, as well as those governments that held consultations to directly inform the authoritative strategy document.

The authors considered only those strategies embodied in a single, authoritative document. If a country held a consultation for an element that might serve as part of its overall "strategy" for AI, but not for its national strategy document, it was not counted. For example, Colombia produced multiple documents and initiatives, rather than relying on one unifying document for its national AI strategy. Thus, Colombia was not counted in the analysis (Republica de Colombia Departamento Nacional de Planeación 2019).

It should be noted that AI strategies are living documents and governments at times update such strategies. Once the authors determined which documents were to be included as AI strategies and which countries had public consultations (Australia, Brazil, Chile, France, Germany, India, Indonesia, Ireland, Italy, Jordan, Malaysia, Norway, Peru, Poland, Türkiye, the United Kingdom and Uruguay), the authors began to focus on defining the research questions (see Box 2 below).

Box 2: Research Questions

1. How and when did the government engage with its citizens in the creation and adoption of the national AI strategy?
2. What materials did the government provide to prepare/enable the public to give informed advice on the AI strategy?
3. Did the government make efforts to ensure a broad cross-section of people knew about and could comment on the strategy?
4. Who participated in the engagement processes?
5. Did the government provide evidence that it made use of the feedback it received?

Next, data was gathered on these 18 cases of national AI strategies that had a public consultation. With such data, the authors were able to develop 14 indicators to use to characterize answers to the six research questions (see Table 1).

Table 1: Indicators

Research Question	Indicator
1. How and when did the government engage with its citizens in the creation and adoption of the national AI strategy?	Was there a formal government consultation open to the unorganized public?
	Did the government consult a closed group of experts, at least one of whom was a representative of a civil society group?
	Did the government engage the public in settings outside the formal consultation, for example, workshops or round tables?
	Was there public input on the initial stage of the development of the AI strategy?
	Was there public engagement prior to the release of the final/official strategy?
	Were there multiple stages of public engagement, often resulting in the publication of interim or draft documents before the official strategy?
2. What materials did the government provide to prepare/enable the public to give informed advice on the AI strategy?	Did the government provide any relevant background documents in its formal consultation(s)?

	Did the government provide adequate information to participants of other engagement mechanisms?
3. Did the government make efforts to ensure a broad cross-section of people knew about and could comment on the strategy?	Were there both online and offline options for participation?
	Were there any efforts to promote the inclusion of communities facing specific difficulties, such as the disabled, or to promote the inclusion of historically marginalized groups?
4. Who participated in the engagement processes?	Was it possible to identify participants by seeing the comments and/or who made them?
	Did the government release a summary or other report detailing the comments that included information about the makeup of the respondents?
5. Did the government provide evidence it made use of with the feedback it received?	Did the government acknowledge the comments it received, either in the strategy itself or elsewhere?
6. Did any of the consultations achieve the IAP2 level of “Involve”?	Did the government explain how it incorporated comments into the strategy?

To supplement the data gathering, the authors contacted the various agencies and individuals responsible for the strategy’s development. However, despite this feedback and reliance on public information, their analysis may include incomplete information.

Finally, the authors condensed the answers to the indicators for each of the 18 countries into answers for each research question. Next, that information was synthesized into the case studies in the Annex. No assessment was made on whether these countries had an effective consultation. The findings and background data will be available at the Digital Trade and Data Governance Hub research website, under Public Participation in AI Strategies.

Findings

The six research questions enabled an assessment of whether, how, when, who and to what extent nations consulted with their public on their AI strategy (see Box 2¹). In this section, the authors summarize answers to each research question are summarized. For country-specific information, please see the Annex.

1. How and When Did the Government Engage with Its Citizens in the Creation and Adoption of the National AI Strategy?

The authors found significant variation in how and when nations engaged with their citizens on their AI strategies. Twelve nations from the sample of 18 adopted a similar process in developing their AI strategy. They began by convening experts from business, professional associations, government and professional associations. Six of these nations (Chile, Indonesia, Italy, Peru, the United Kingdom and Uruguay) convened a working group or expert committee while another six (Australia, Brazil, France, Germany, Poland and Türkiye) consulted individual experts. They then expanded the circle of those consulted in the hope of receiving comments from a wide range of citizens (OECD 2022). Following this expert input, Australia, Brazil, Chile, Germany, Italy, the United Kingdom, Uruguay and Poland then produced a draft strategy or a discussion paper focused on what a strategy could include.

Six other nations in the sample took a different approach. Türkiye consulted experts and then released the official strategy; India, Indonesia and Peru released what they called a “draft” strategy that serves as their official strategy. France, Ireland, Malaysia and Norway did not produce a working document, but held public consultations before releasing the official strategy. Jordan announced its official strategy after the analysis was finished.¹⁸

Some 13 nations (Australia, Brazil, Chile, France, Ireland, Italy, Jordan, Norway, Peru, Poland, the United Kingdom, the United States and Uruguay) held formal public consultations at some point in the process of developing a strategy. Interestingly, these nations differed in how and when they sought public comment. Some governments obtained public input through a survey; others requested comments on the draft strategy or discussion paper produced by the expert group; others asked for comments on broad regulatory issues; or, in Norway's case, simply issued an open invitation to submit input on the subject (Norway Ministry of Local Government and Modernisation 2019). Some of the 13 also organized workshops, focus groups and round tables. Seven of these governments claimed they held such additional events (Australia, Chile, Ireland, Indonesia, Malaysia, Türkiye and the United Kingdom), but in three cases (Ireland, Indonesia, Türkiye) no mention of them could be found outside the strategy text. Malaysia held virtual town halls

¹⁸ See United Nations Industrial Development Organization (2022).

and virtual focus group discussions (Jamil 2021), and Chile, in addition to its release of a tentative index for comment, held numerous regional and self-convened workshops around the country.¹⁹ Some nations also held conferences (Australia, Malaysia, Poland), webinars (Chile, Poland), and other events with partners (the United Kingdom), to inform the general public about the AI strategy and its development. Norway followed its open call for comments from the public with a series of in-person meetings between the minister of digitization and various stakeholder groups. Italy held two public consultations on separate draft strategies (Canna 2021). Uruguay asked for public comment on principles to guide the strategy, then on the draft strategy (Hernández and de Souza 2022, 27).

For policymakers in several nations, obtaining public comments was an ongoing process. For example, Malaysia described its AI road map as a “living document” that will be continually updated based on further feedback (Ministry of Science, Technology & Innovation Malaysia 2021), and Peru’s strategy calls for updates every two years.²⁰ Malaysia and Indonesia host their strategies on webpages where individuals can still provide comments (as of January 2023).²¹ Germany and the United States have subsequently released updated strategies,²² and India and the United Kingdom have released implementation or guidance documents.²³ All four nations that published additional documents after the release of the national strategy consulted the public in some form in advance of these updates (although Germany again only consulted organizations).

2. What Materials Did the Government Provide to Prepare/Enable the Public to Give Informed Advice on the AI Strategy?

The governments in the sample provided several different types of documents to assist their constituents in providing comments on the AI strategy. Some countries gave their citizens a draft strategy or a preliminary document prepared by either the expert committee or the relevant government agency. Five nations (Germany, Italy, Poland, the United Kingdom and Uruguay) provided a document delineating recommendations from experts. Chile, Italy, Poland and Uruguay provided respondents with a draft strategy developed for the purpose of the consultation. In contrast, India, Indonesia, Jordan and Peru released the official, final strategy and asked for public comment at that point,

¹⁹ See Ministerio de Ciencia, Tecnología, Conocimiento e Innovación, Políticas/Política_Inteligencia_Artificial/Mesas_Regionales/. https://github.com/MinCiencia/Políticas/tree/main/Política_Inteligencia_Artificial/Mesas_Regionales.

²⁰ See <https://cdn.www.gob.pe/uploads/document/file/1909267/National%20Artificial%20Intelligence%20Strategy%20-%20Peru.pdf>.

²¹ See <https://ai-innovation.id/strategi> and for Malaysia, see Canna (2021).

²² See www.ki-strategie-deutschland.de/files/downloads/201201_Fortschreibung_KI-Strategie.pdf and www.nitrd.gov/pubs/National-AI-RD-Strategy-2019.pdf.

²³ See www.niti.gov.in/sites/default/files/2021-02/Responsible-AI-22022021.pdf and www.gov.uk/government/publications/national-ai-strategy-ai-action-plan.

without an intervening document or additional information that could help citizens understand the relevant issues. Australia released a discussion paper, which calls for responses to questions rather than any preconceived recommendations (Australian Government Department of Industry, Science, Energy and Resources 2020). Brazil presented its public with a description of predetermined thematic pillars based on OECD recommendations, around which the strategy would be designed, with discussion questions (Ministério da Ciência, Tecnologia, Inovações e Comunicações 2019). Similarly, France sought comments on briefly described thematic courses of actions, and also explicitly sought proposals for additional courses of action and discussion between commenters.²⁴ In contrast, although the United States twice updated its AI strategy, it did not provide additional material to its constituents. Individuals could, however, review the 2016 strategy to provide comments on the 2019 revision or on the 2019 revision to provide comments on the 2022 update (Office of Science and Technology Policy 2022). Finally, Ireland provided very little information to its public, including only short descriptions of the strategy's objectives (Department of Business Enterprise and Innovation 2019).

To summarize, the governments that held formal public consultations provided background material on guiding principles, objectives and/or the strategy itself. However, these documents could not enable the broad public to give well-informed comments on AI. Most people need to understand more about AI systems and their risks and benefits to comment effectively on an AI strategy, but few governments made the effort to prepare their citizens to effectively comment. As an example of what governments could do, Finland created a free online course to demystify AI, in recognition that its citizens must trust when governments or other entities use AI;²⁵ however, Finland did not consult with its citizens on its AI strategy.

3. Did the Government Make Efforts to Ensure a Broad Cross-Section of People Knew About and Could Comment on the Strategy?

Government officials relied on government websites or emailed surveys to request public opinion on their AI strategy. To participate, one therefore needed internet access, which is not available or affordable for all people. Some governments took additional steps to broaden the circle of commenters, although in almost all cases these, too, were online-only. As an example, Chile organized regional round tables to get feedback from people throughout the country; these were all online except for the first two, but if it was not for the acute impact of the pandemic forcing the organizers to change their plan, more would

²⁴ See <https://purpoz.com/project/mettre-en-place-un-terreau-general-favorable-au-developpement-de-l-ia/consultation/consultation-32/consultations>.

²⁵ See www.elementsofai.com/.

have been in person.²⁶ In Malaysia, the government organized a virtual conference, focus group discussions and a virtual town hall to get more people involved in the consultations.

The authors could not ascertain whether the governments were successful at attracting diverse comments from a wide range of their constituents. No information on the numbers from India, Italy, Peru, Poland and Jordan could be found. (Jordan finished its consultation as this paper was being finished.)

Table 2 reveals that few people actually participated in public consultations in most of the sample. It could not be determined whether the low number of participants stemmed from a lack of interest among constituents or whether it stemmed from government unwillingness or failure to extensively market their outreach efforts.

Table 2: Total Number of Comments and Commenters in Public Consultation

Country	Number of Comments and Commenters
Australia	90 submissions; no information on participants
Brazil	1,000 contributions; no information on the number of participants
Chile	209 (unclear whether comments or commenters)
France	1,639 "attendees"; 2,407 contributions
Germany	88 unique responses (Zablecalculation)
India	No information available
Indonesia	57 unique responses (Zablecalculation)
Ireland	92 survey responses
Italy	No information available
Jordan	No information (yet) available
Malaysia	173 survey responses
Norway	51 unique responses (Zablecalculation)

²⁶ See www3.bcn.cl/observatorio/asiapacifico/noticias/jose-guridi-borrador-politica-nacional-IA.

Peru	None found for consultation survey
Poland	No information available
Türkiye	206 interviews, 108 workshop participants
United Kingdom	413 survey responses
United States	46 unique responses (Zable calculation)
Uruguay	28 comments, but the website was taken down so a careful review was not possible

Source: Adam Zable analysis

4. Who Participated in the Engagement Processes?

The authors struggled to answer this question because many in the sample did not provide such information or provided only some information about participation. In most instances, the government provided only a person's name or name and job, thus, it was not possible to categorize the participants as either members of the general (unorganized public) or organized public.

Three countries clearly provided information on the background of individuals that participated in consultations on the AI strategy, and it was possible to ascertain the background of others from websites (Brazil, Chile, France, Germany, Indonesia, Ireland, Malaysia, Norway, Türkiye, the United Kingdom, the United States and Uruguay). Chile²⁷ and the United Kingdom (Alan Turing Institute 2021) delineated their participants' gender, geographical region and profession or association, so it was easy to ascertain who participated. Malaysia (Ministry of Science, Technology & Innovation 2021) and Türkiye (Republic of Türkiye Ministry of Industry and Technology 2021) described participants by the category of their professional institution (NGO/government/business/academia). Brazil,²⁸ Norway²⁹ and the United States³⁰ maintain websites where the consultation was hosted and on which one can see both the comments and who commented, in a form that allows for subsequent analysis. Uruguay took its website down during the course of this research, but comments and commenters were previously viewable.

²⁷ See https://minciencia.gob.cl/uploads/filer_public/6c/c1/6cc17cd7-ae58-48fo-ada1-d33a3e6e8958/informe_consulta_publica_ia_1.pdf.

²⁸ See www.gov.br/mcti/pt-br/acompanhe-o-mcti/transformacaodigital/arquivosinteligenciaartificial/ebia-consulta-publica.pdf

²⁹ See www.regjeringen.no/no/tema/statlig-forvaltning/ikt-politikk/KI-strategi/mottatte-innspill-til-ki-strategien/id2640057/

³⁰ See www.nitrd.gov/coordination-areas/ai/ai-rfi-responses-2018/

It was difficult to compare information on participants among countries. Moreover, Australia, France, India, Indonesia, Italy, Jordan, Peru and Poland did not provide any information on who participated in their consultations. For example, Australia gave the number of respondents with no further information,³¹ while France and Indonesia provided comments and the names of the participant, which made it difficult to assess the background of the participant. Table 3 describes what was found for the 18-country sample.

Table 3: Breakdown of Who Commented on National AI Strategies

Country	Who Commented
Australia	No information available.
Brazil	31 commenters: • private sector\professional organization — 45% • NGO/third sector — 19% • government — 3% • academia — 19% • unaffiliated individual — 13%
Chile	Results report breaks 209 participants down by age, country, gender, educational level and geographical location within Chile, although not by professional sector. 86.31% were natural persons and 13.9% legal persons. There is a breakdown of the participants of the regional workshops by professional sector on the online platform that hosts the consultation details.
France	Consultation summary report breaks down comments as follows: • 1,639 attendees • 2,407 contributions While comments are available for viewing on the online portal, commenters' names were often not given in full, preventing an assessment of the makeup of respondents.
Germany	No members of the unorganized public were involved. Of the organizations, the analysis shows the following, for the 2018 consultation: • 88 commenters • private sector — 52.2% • professional organization — 27.2% • NGO — 11.3% • government — 3.7% • academia — 5.6%

³¹ Ministerio de Ciencia, Tecnología, Conocimiento e Innovación, Política Nacional de Inteligencia Artificial. https://minciencia.gob.cl/uploads/filer_public/bc/38/bc389daf-4514-4306-867c-760ae7686e2c/documento_politica_ia_digital.pdf

India	No information available.
Indonesia	While comments are available for viewing on the online platform, commenters' names were all that was given, preventing an assessment of the makeup of the 57 respondents.
	The public consultation report breaks down the 85 commenters as follows: • businesses — 16% • government — 5% • research/academia — 46% • representative body — 7% • anonymous — 12% • general public — 14%
Ireland	In addition, seven primarily business groups provided written statements separately.
Italy	No information available.
Jordan	No information yet available.
Malaysia	According to the road map, the government received 173 responses to the survey, which it breaks down as follows: • industry/private/companies — 45% • government — 38% • academia — 14% • other — 3%
Norway	According to the analysis of the 51 comments: • business/business association — 20.7% • government — 20.7% • academia — 15% • professional association — 26.4% • NGO/civil society/charity/third sector — 13.5% • non-affiliated individual — 3.7%
Peru	No information available.
Poland	No information available.
Türkiye	According to the strategy, 36 interviews were held with government representatives, 38 with the private sector, three with NGOs, 26 with universities and 103 with domain experts in different disciplines. The strategy also says that two workshops were held that were attended by 40 representatives from public institutions, 38 from private sector organizations, 26 from academia and four from NGOs.
United Kingdom	<i>AI Ecosystem Survey Results Report</i> breaks down the 413 respondents as follows: industry (44%) and academia (32%), with a large minority from the public sector (15%).

	Further questions include information about professional background, geographical region of work, position at work, age, gender and ethnic group.
United States	According to the analysis of the 46 respondents: • business/business association — 36.9% • government — 0% • academia — 11% • professional association — 30.1% • NGO/civil society/charity/third sector — 13% • non-affiliated individual — 9%
Uruguay	The webpage that hosted the consultations was taken down during the course of this research, preventing an analysis of the 28 comments.

Source: Adam Zable analysis

5. Did the Government Provide Evidence That It Made Use of the Feedback It Received?

Public feedback is an essential element of good governance, and thus many governments seek comments to improve public policy and to be responsive to their citizens (OECD 2011). Hence, it was important to examine if a government indicated whether, and if so how, it made use of the comments given during the course of the consultations. Only four countries provided concrete evidence that they incorporated such comments:

- In Chile, the government said it incorporated learnings from workshops (Ministerio de Ciencia, Tecnología, Conocimiento e Innovación 2020), and a post-consultation results report discussed how the government incorporated suggestions in the strategy (Ministerio de Ciencia, Tecnología, Conocimiento e Innovación 2021). Officials modified and refined the content and wording of objectives, and added entirely new objectives and strategies based on the contributions received, including new strategies to promote the regulation of personal data, launch initiatives in priority areas and foster communities of users of public interest data.
- In the United States, the government explained that based on feedback it received, officials added a new strategy to expand public-private engagement (Select Committee on Artificial Intelligence 2019).
- The German government did not explicitly say how it used comments, but Germany's AI strategy dedicated substantial space to summarizing and explaining the comments, suggesting that the comments impacted the direction of the strategy.
- A Uruguayan official replied to each comment posted on the consultation portal describing how the government would incorporate **these comments**.³²

³² See www.gub.uy/participacionciudadana/consultapublica/legislation_proposals/3-consulta-publica-propuesta-de-la-estrategia-de-inteligencia-artificial-para-el-gobierno-digital&cd=2&hl=en&ct=clnk&gl=nl.

- In France and Malaysia, the strategy text summarized or gave reference to comments the organizers received but did not indicate how it responded to specific comments.

6. Did Any of the Consultations Achieve the IAP2 Level of “Involve”?

The IAP2's Spectrum of Political Participation (Figure 1) is designed to illustrate different modes of participation. The authors relied on this metric to characterize whether any nation's engagement strategy for the development of their national AI strategy went beyond simple consultations to involvement, consulting the public, to more actively involving the public in the process.

Figure 1: IAP2 Spectrum of Public Participation



IAP2 Spectrum of Public Participation

IAP2's Spectrum of Public Participation was designed to assist with the selection of the level of participation that defines the public's role in any public participation process. The Spectrum is used internationally, and it is found in public participation plans around the world.

INCREASING IMPACT ON THE DECISION 					
	INFORM	CONSULT	INVOLVE	COLLABORATE	EMPOWER
PUBLIC PARTICIPATION GOAL	To provide the public with balanced and objective information to assist them in understanding the problem, alternatives, opportunities and/or solutions.	To obtain public feedback on analysis, alternatives and/or decisions.	To work directly with the public throughout the process to ensure that public concerns and aspirations are consistently understood and considered.	To partner with the public in each aspect of the decision including the development of alternatives and the identification of the preferred solution.	To place final decision making in the hands of the public.
PROMISE TO THE PUBLIC	We will keep you informed.	We will keep you informed, listen to and acknowledge concerns and aspirations, and provide feedback on how public input influenced the decision.	We will work with you to ensure that your concerns and aspirations are directly reflected in the alternatives developed and provide feedback on how public input influenced the decision.	We will look to you for advice and innovation in formulating solutions and incorporate your advice and recommendations into the decisions to the maximum extent possible.	We will implement what you decide.

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Source: © International Association for Public Participation www.iap2.org.

To make this useful for the categorization of public participation, the IAP2 categories were expanded into specific indicators:

- Inform: Did the government provide the public with information to assist them in understanding the issue before the strategy was published?
- Consult: Did the government ask for comment from the organized and unorganized publics?
- Acknowledge: Did the government acknowledge public comment?

- Respond: Did the government provide feedback on how public input influenced the final strategy?
- Involve: Did the government work directly with the public throughout the process?
- Collaborate: Did the government work with the public to develop the initial direction, and provide evidence that it incorporated public concerns into the decisions made?
- Empower: Did the government place final decision-making authority in the hands of the public?

The findings are provided in Table 4 below.

Table 4: Adapted IAP2 Spectrum

	Inform	Consult	Acknowledge	Respond	Involve	Collaborate	Empower
Australia	☒	☒	☒	☐	☐	☐	☐
Brazil	☒	☒	☒	☐	☐	☐	☐
Chile	☒	☒	☒	☒	☒	☐	☐
France	☒	☒	☒	☐	☐	☐	☐
Germany	☒	☐	☒	☒	☐	☐	☐
India	☐	☒	☐	☐	☐	☐	☐
Indonesia	☐	☒	☐	☐	☐	☐	☐
Ireland	☒	☒	☒	☐	☐	☐	☐
Italy	☒	☒	☐	☐	☐	☐	☐
Jordan	☐	☒	☐	☐	☐	☐	☐
Malaysia	☐	☒	☒	☐	☐	☐	☐
Norway	☐	☒	☒	☐	☐	☐	☐
Peru	☐	☒	☐	☐	☐	☐	☐
Poland	☒	☒	☐	☐	☐	☐	☐
Turkey	☐	☐	☒	☐	☐	☐	☐
United Kingdom	☒	☒	☒	☐	☐	☐	☐
United States	☒	☒	☒	☒	☐	☐	☐
Uruguay	☒	☒	☒	☒	☐	☐	☐

Source: Authors.

As the table shows, most of the countries that developed an AI strategy barely involved their citizens. Seven of the 18 nations fall short of achieving even the “Inform” level in engaging their publics, as they either sought public input after finalizing their AI strategies or failed to provide adequate contextual information. In India, Indonesia, Jordan and Peru, the government asked for public comment only after the strategy was fully written and released. Norway provided no information when it asked for public comment, and it was

not possible to verify whether information had been provided prior to public engagement in Türkiye and Malaysia.

Next, the researchers examined if nations went beyond informing their citizens to consulting them. Consultations can, , vary in their degree of openness: they can be accessible to anyone who wants to comment or closed to only certain sectors of society (OECD 2011). In the cases of Germany³³ and Türkiye, the government directed the consultation mechanisms at the *organized* public, meaning stakeholder organizations (including civil society organizations). However, they did not provide a means by which the *unorganized* public could comment.

The researchers then examined whether case study governments acknowledged the comments they received. Six countries did not acknowledge receiving public comments. In some countries, such as India and Peru, the process is ongoing and these governments could acknowledge and incorporate comments received in past consultations in future updates.³⁴ Moreover, only Chile, Germany, the United States and Uruguay indicated how they changed their strategy in response to public comments. France, Ireland, Malaysia, Türkiye and the United Kingdom summarized the comments they received in graphs, statistics or text; included individual comments in the final strategy; and/or provided the comments themselves or the individuals and organizations who commented. However, such summaries and statistics are not evidence that the government was responsive and accountable to these commentors.

Finally, , only Chile reached the point of “Involve.”³⁵ Chile’s engagement strategy involved multiple stages of progressively widening participation by various publics, starting with a small group of experts, then engaging the wider public through the release of a preliminary index, which was followed by webinars, workshops and a formal public consultation. Moreover, Chilean officials were responsive to public comment throughout the process. Chile also maintains perhaps the best record of these mechanisms, with a GitHub page that collects most of the relevant information, which is still accessible as of November 2022.³⁶ Taken in sum, while most governments informed and some consulted some of their constituents, the authors concluded that no country “collaborated” with its citizens according to the IAP2 metric.

³³ See www.ki-strategie-deutschland.de/.

³⁴ <https://www.unido.org/news/jordan-presents-its-ai-strategy-and-implementation> for artificial intelligence and the implementation plan (2023-2027) for public consultation, 2022.

³⁵ Ireland and Malaysia stated that they held workshops, round tables, and other meetings, but evidence of these outreach efforts could not be found despite attempts to contact the responsible agencies.

³⁶ See https://github.com/MinCiencia/Politic/tree/main/Politica_Inteligencia_Artificial/Mesas_Regionales

Conclusion

Most governments want to build trust in AI given the importance of AI to national security and their current and future economic growth. Yet their strategies to encourage AI are unlikely to build and sustain that trust because policy makers have not sought to inform, consult, involve and collaborate with many of the same people who might be affected by the misuse of AI.

The authors examined how and when governments sought public comments on their AI strategies, as well as who participated. The sample included 68 countries and the European Union. Of the 43 with AI strategies, 25 did not consult the public. To put it differently, these 25 nations missed an opportunity to build trust. Only 18 nations in the sample tried to obtain feedback from members of the public.

Many of the cases utilized a similar set of actions to obtain public comment. The process often began when policy makers set up an advisory group or consulted directly with experts from business and academia. These groups have the understanding and experience articulating their concerns and priorities towards AI. Next, the government built on the expert consultation to produce a draft AI strategy or other preliminary document. Often policy makers used that draft strategy to obtain comments. Then the government announced its AI strategy. Most governments, however, did not describe how they incorporated public comments.

Some governments created an ongoing consultative process that was embedded within a more extensive (ongoing) agenda of AI governance. For example, Germany, the United States, India and the United Kingdom have released additional AI strategic governance documents, all of which contained some element of public input, and Peru and Malaysia intend to do the same. However, the process could be delayed or abandoned with a change in government.

Unfortunately, most governments did little to facilitate informed comments by their citizens. They did not explain how AI might affect their constituents in their daily lives or in their many roles as citizens, producers, consumers, and advocates. Moreover, they did not explain the benefits and risks of AI to individuals and society as a whole. Certainly democratic governments should do more and encourage their allies to broaden this discussion.

Governments also made little effort to get the word out to their constituents and to motivate them to participate in developing the strategies. In general, they used websites and online platforms to inform their citizens about the consultation. But these policy

makers could do more. The OECD noted that when governments seek consultation, they should use a wide range of outreach methods, such as advertising, video primers, partnering with civil society groups or educational institutions. They could also provide economic incentives, as several surveyors or pollsters do (*ibid.*, 29, 31). Policy makers' failure to do so raises the question as to whether they really wanted such comments.

Because they failed to attract significant public input, policy makers generally relied on the recommendations of experts to guide public input. While it was possible to see who commented in 12 nations, very few governments provided detailed breakdowns of the participants. Consequently, it is not possible to say whether the participants were truly representative of the unorganized public or mainly representatives of the organized public. Moreover, the numbers of people commenting were relatively small and may not be representative of the nation as a whole.

Most of the sample did little to involve historically marginalized groups that could be significantly affected by government and private-sector use of AI systems (Hernández and de Souza 2022, 27.) Most governments provide information in their national language and not in Indigenous and foreign languages, which could make it harder for some of their citizens (or foreigners) to participate. In their report, *Derechos Digitales* suggested that policy makers could broaden public participation by translating informational material into local languages and making special efforts to involve communities that lack internet access. Moreover, they could, for example, hold “in-person meetings for people with disabilities or geographically distant communities” (*ibid.*, 12).

Some nations, such as Chile, hosted in-person events, but most governments relied solely on online portals or websites. These nations did not provide a means for those without internet access to comment. Moreover, most nations did not provide a long period of time to provide comments — it averaged around a month in the 18-country sample. Although Brazil allowed comments for three months, some countries gave as little as two weeks, which would limit the number of participants.

In conclusion, policy makers are missing an opportunity to build understanding of AI, sustain trust in AI and build a more participatory process for governing AI. Most nations have done little to prepare and involve their citizens for this discussion. At the same time, most people lack the ability, information and resources to participate meaningfully in AI governance. If they want to build greater trust in AI, policy makers should also find ways to incentivize broader participation and to ensure that regulators, legislators and AI deployers truly listen to what their citizens have to say. Without the input of a wide swathe of their citizenry, policy makers may struggle to anticipate future problems related to AI and, over time, to sustain trust in AI systems.

Finally, it should be noted that AI systems are often deployed globally, yet many people in the developing world lack a voice in determining how and whether these systems are utilized to make decisions, predictions or recommendations. Policy makers around the world should find ways to seek out and incorporate their perspective. **Works Cited**

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